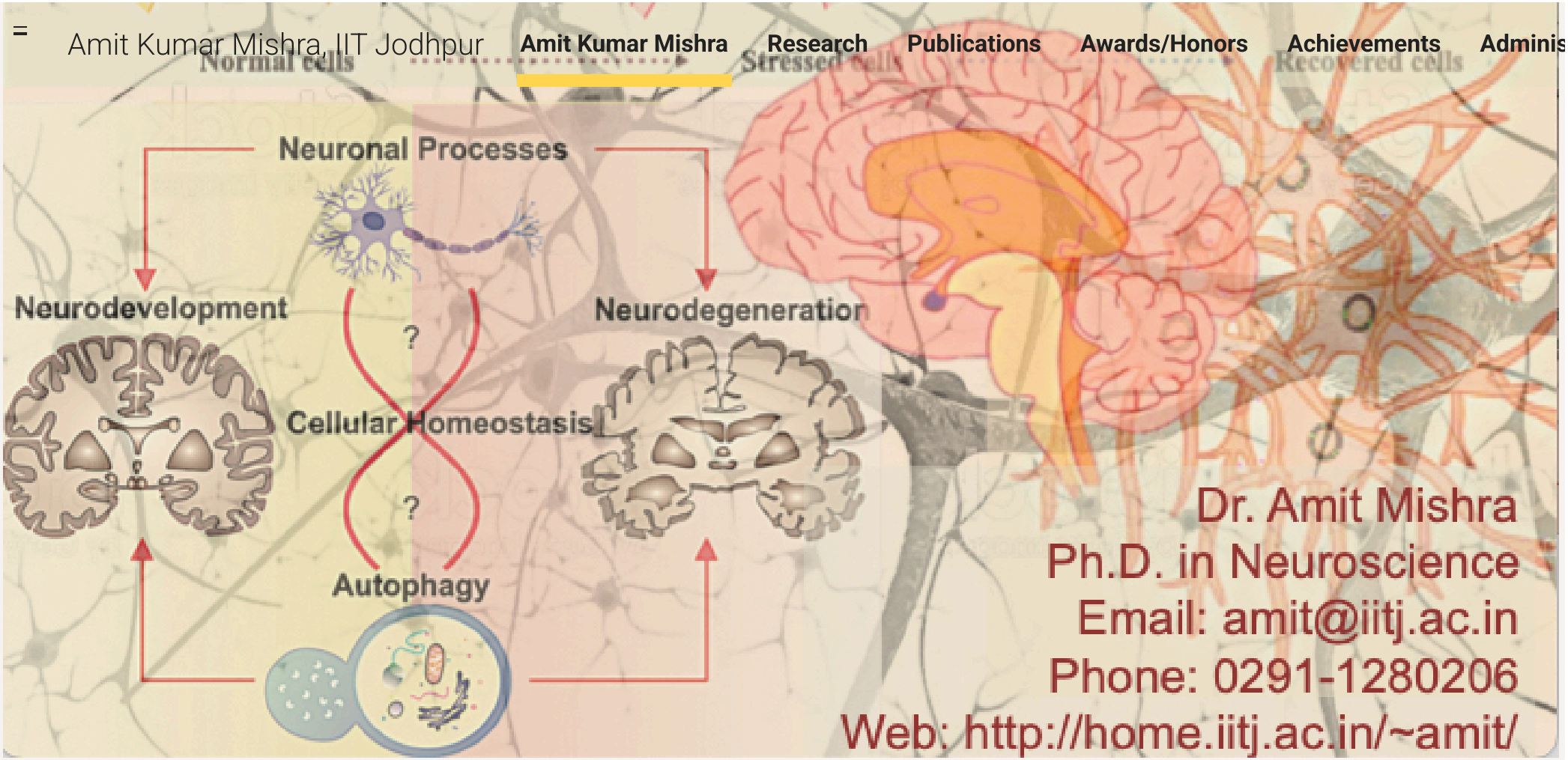
Amit Kumar Mishra, IIT Jodhpur

TARSONS
BECKMAN COULTER

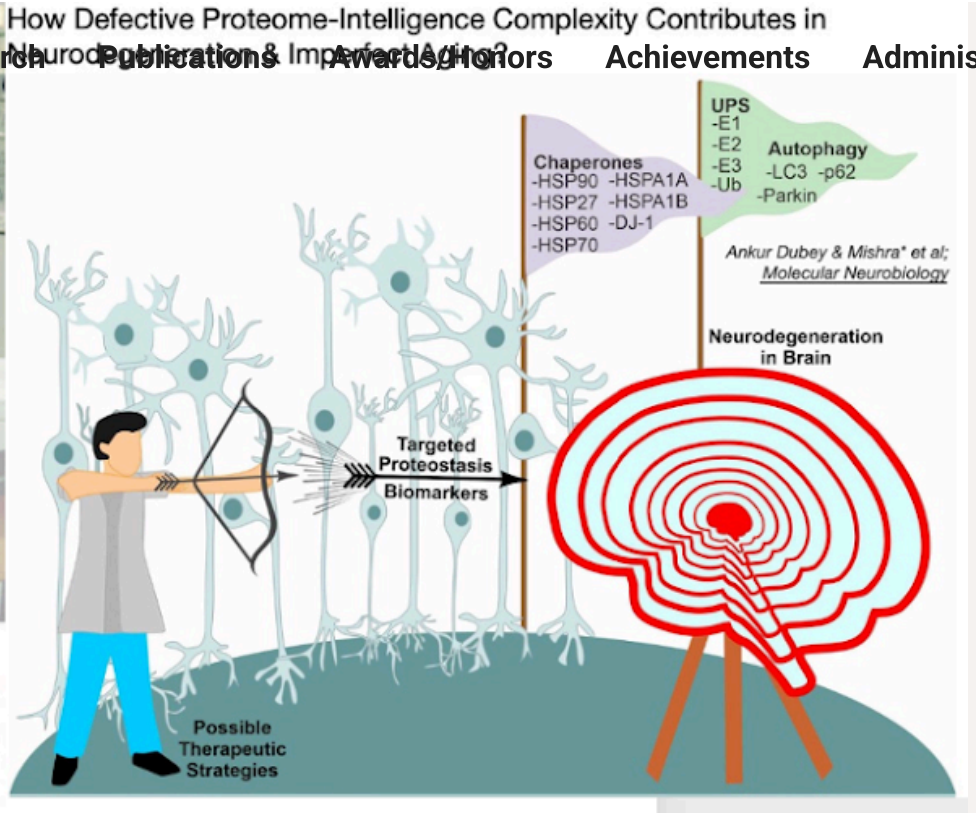

Normal cells
Stressed cells
Recovered cells
Neuronal Processes
Neurodevelopment
Neurodegeneration
?
Cellular Homeostasis
?
Autophagy
Dr. Amit Mishra
Ph.D. in Neuroscience
Email: amit@iitj.ac.in
Phone: 0291-1280206
Web: http://home.iitj.ac.in/~amit/

Amit Kumar Mishra completed his **M.Sc. in Biotechnology with honors** from Devi Ahilya University, Indore. He then joined Ph.D. in Neuroscience at Cellular and Molecular Neuroscience Unit, under the supervision of Prof. Nihar Ranjan Jana (Current Position: Professor, Indian Institute of Technology Kharagpur) from **National Brain Research Centre, India** (Established by Padma Sri, Padma Bhushan and Padma Vibhuhassan Prof. P.N Tandon and Shanti Swarup Bhatnagar Prize, Padma Shri Prof. Vijaylakshmi Rabindranath). Dr. Mishra, Ph.D. The thesis entitled "Understanding the molecular function of E6-AP-a putative E3 ubiquitin ligase mutated in Angelman mental Retardation Syndrome" was selected for **BEST Ph.D. THESIS AWARD in Biological Sciences from the Indian Society of Chemists and Biologists, India, and also awarded by the Indian Academy of Neuroscience for the Best Presentation Award (IST Prize).** He investigated the role of key players in protein quality control mechanisms: E3 ubiquitin ligase and molecular chaperones. Understand the role of E6-AP E3 Ubiquitin Ligase in p53 (tumor suppressor protein) proteasomal degradation.

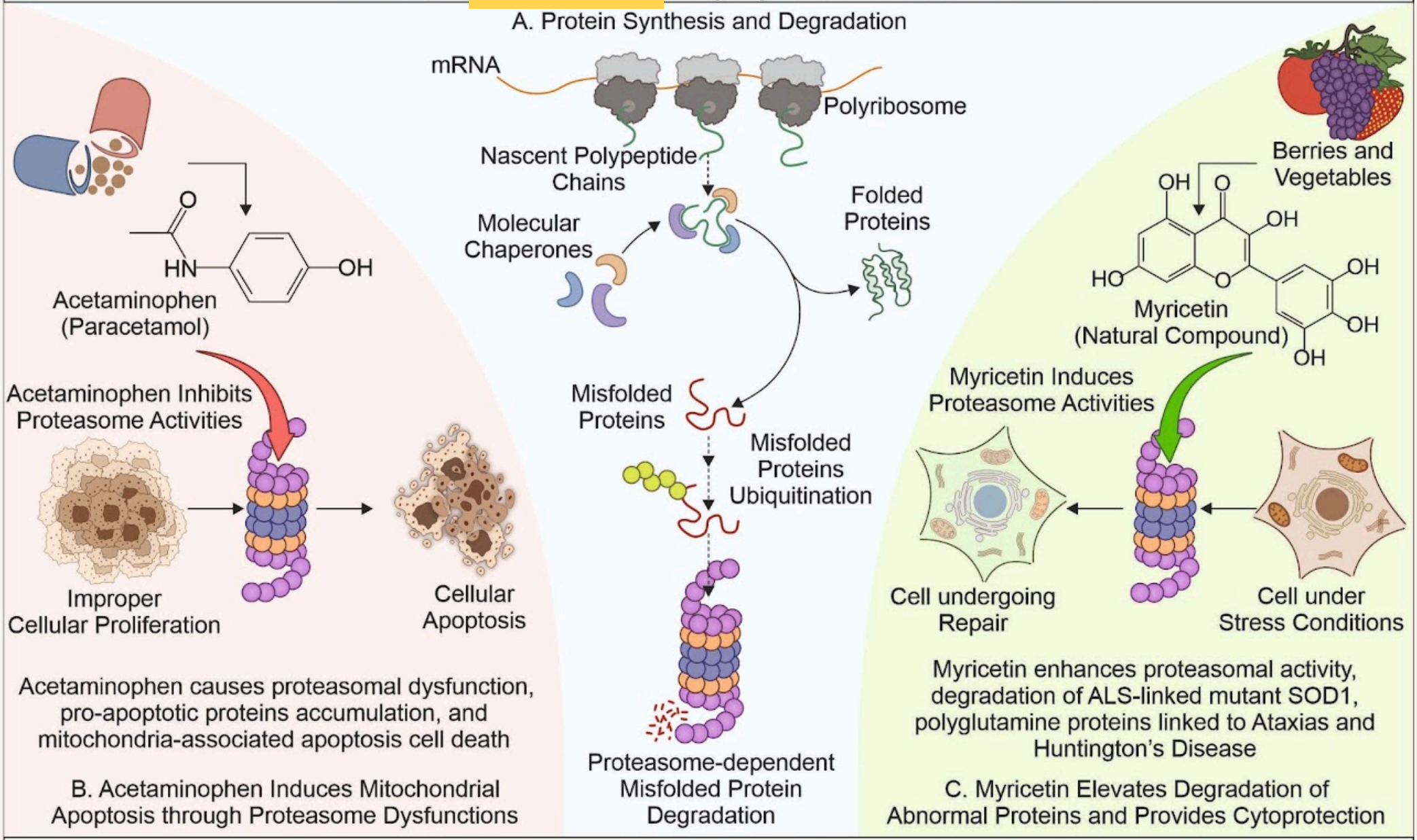
He also observed that p27/Kip1 (Cyclin-dependent kinase inhibitor) ubiquitination and degradation are regulated by E6-AP E3 Ubiquitin Ligase through proteasomal degradation. The major focus of Amit's research projects has been the study of "Molecular Pathogenesis Mechanism of Various E3 Ubiquitin Ligases and Chaperones Implicated in Neurodegenerative Diseases, Neurodevelopmental Disorders, and Ageing". **The New York Academy of Sciences (the Oldest Science Academy in the World), with members from over 100 countries around the globe, has recognized Amit's Profile for exceptional research in Neurodegeneration.** In his NYAS profile, Amit defines science as "an art with measurements that provides new thinking, unlimited imagination, and a solid platform to discover hidden truths on our planet". Despite noteworthy progress in our current understanding, maintenance of functional proteome, or proteostasis, is critical to delaying imperfect aging-like processes. However, many lines of evidence suggest that proteostasis failure may also lead to fatal.

Current Positions:
Professor, IIT Jodhpur
Fellow of Royal Society of Medicine
Fellow of Royal Society of Biology
Bicentennial Ambassador (NYAS)
Member of National Academy of Science India
Member of National Academy of Medical Science
Member of Royal Society of Chemistry

Prof. Amit Mishra
Indian Institute of Technology Jodhpur
Ph.D. in Neuroscience
National Brain Research Centre
RIKEN Brain Science Institute Japan
Max Planck Institute of Biochemistry, Germany



Proteasome Functions Based Innovative Molecular Strategies Against Aberrant Proteins Aggregation and Improper Cellular Proliferation



(A) Proteins are necessary for normal cellular physiology and cells continuously synthesize new proteins to replace old or defective ones. Accumulation of misfolded proteins increases risk for developing various disease conditions. Proteins that cannot fold with assistance of chaperones undergo degradation through Ubiquitin Proteasome System (UPS). (B) Cancer cells utilize cellular proteostasis mechanisms, including UPS,

to evade cell death and promote cellular proliferation. Our study indicates that Acetaminophen induces proteasomal dysfunctions, causing mitochondrial dysfunctions and consequently cell death, highlighting its potential as a pan-proliferative agent. (C) Proteasome defects are also associated with neurodegenerative diseases, and our study shows that Myricetin, a flavonoid, enhances proteasomal activity, facilitating clearance of disease-linked aberrant proteins and reducing their accumulation.

Amit's research precisely focuses on and systematically targets various molecular mechanisms of proteostasis maintenance, as well as discusses progressing neurobiological strategies and promising natural and pharmacological candidates, which can be helpful to counteract the problem of neurodegeneration-associated proteopathies. The quality of work performed by Amit at a young age in India was well proven when he was awarded by **Nobel Laureate-Medicine (Prof. Harald Zur Hausen) and Indian Father of Green Revolution (Prof. M. S. Swaminathan), Fellow Royal Society of London** for his significant contribution in the genetics of molecular Neurodegeneration. His innovative research contribution received the **IYBA Award from Prof. Vijay Raghavan and Honourable Minister Sri Sudini Jaipal Reddy.** Amit's research emphasizes an urgent need to recognize the fundamentals of proteostasis to design a new molecular framework and fruitful strategies to uncover how proteome defects are associated with aging and neurodegenerative diseases. At the international forum, his research contribution was recognized and awarded by **Prof. Venkatraman Ramakrishnan (Nobel Laureate) and Bharat Ratna Prof. C.N.R Rao** in the presence of the Deputy Prime Minister of Singapore under the Royal Society of London Program. His study conclusively demonstrated that few quality-control E3 Ubiquitin Ligases are involved in eliminating misfolded proteins via the selective autophagy pathway.

His entire work was executed in India, and at a young age, he could understand the complex problem of Neurodegeneration. In the presence of Prof. Huda Y Zoghbi from Howard Huges Medical Institute, Baylor College of Medical, he honored the Young Scientist Award at the Japan Neuroscience Society meeting. Further, Mishra's team observed a novel interaction between E3 Ubiquitin Ligases and Molecular Chaperones. He has identified how the cumulative function of E3 Ubiquitin Ligases and Molecular Chaperones additively enhances cytoprotection against neuronal cell death due to oxidative and endoplasmic reticulum stress. Amit secured the Shakuntala Amir Chand Prize of ICMR from **Dr. Soumya Swaminathan, Deputy DG WHO, Prime Minister Scientific Advisor Prof. Vijay Raghavan, and Honourable Minister of Health**. Insufficient or retarded

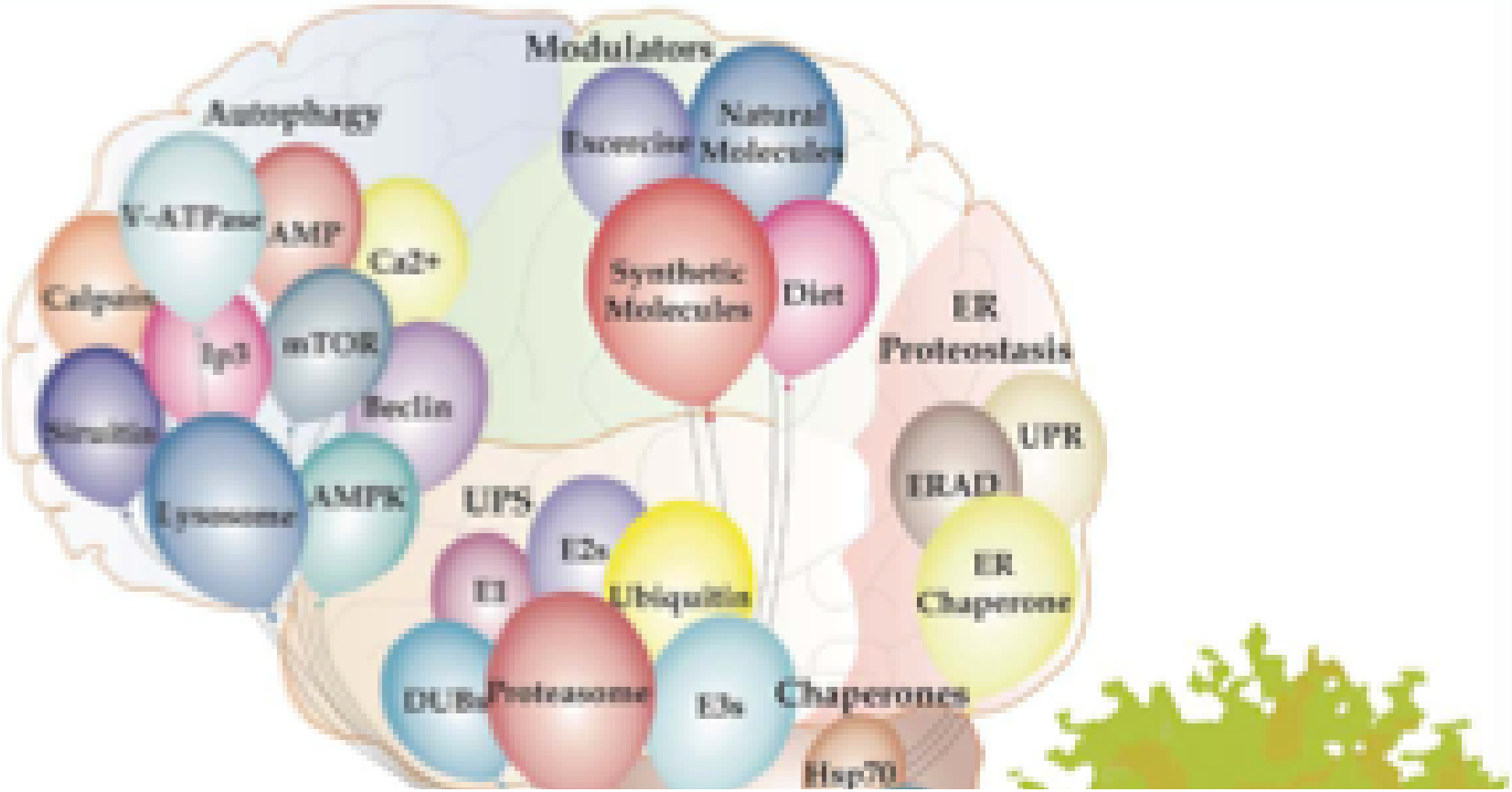
Minister Scientific Advisor Prof. Vijay Raghavan, and Honourable Minister of Health. Insufficient or retarded abnormal protein elimination in neuronal cells leads to various neurodegenerative diseases such as Alzheimer's, Huntington's, Parkinson's, Amyotrophic lateral sclerosis, and polyglutamine diseases. For the first time, Amit's study has elaborated on the quality control function of E3 ubiquitin ligase. He has also been awarded the coveted **Malaviya Memorial Award by Prof. Tej Pal Singh, Prof. Alok Dhawan, Prof. Ashok Pandey, and Prof. Christian Larroche. Amit received the INSA Medal, ISCA Award, and NASI Young Scientist Platinum Jubilee Award.** His research identifies and elaborates the molecular mechanism of protein quality control mechanism, which fails to cause inappropriate folding of proteins.

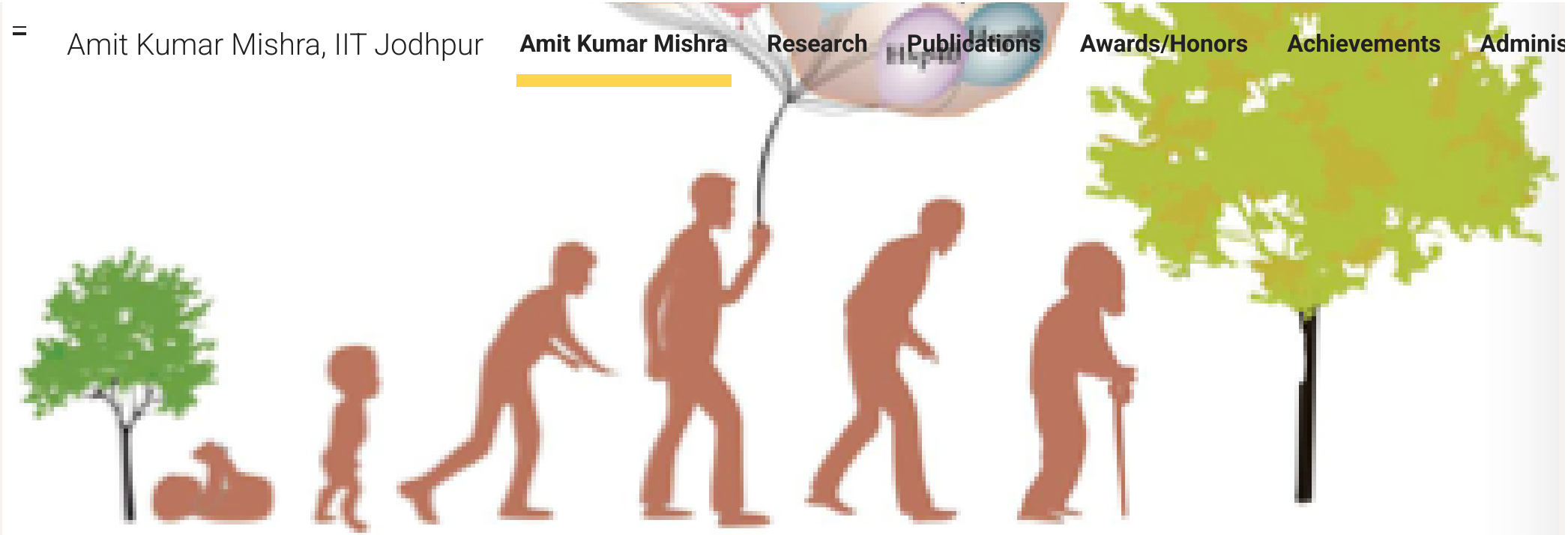
Amit Kumar Mishra, IIT Jodhpur
Amit Kumar Mishra Research Publications Awards/Honors Achievements Admiss

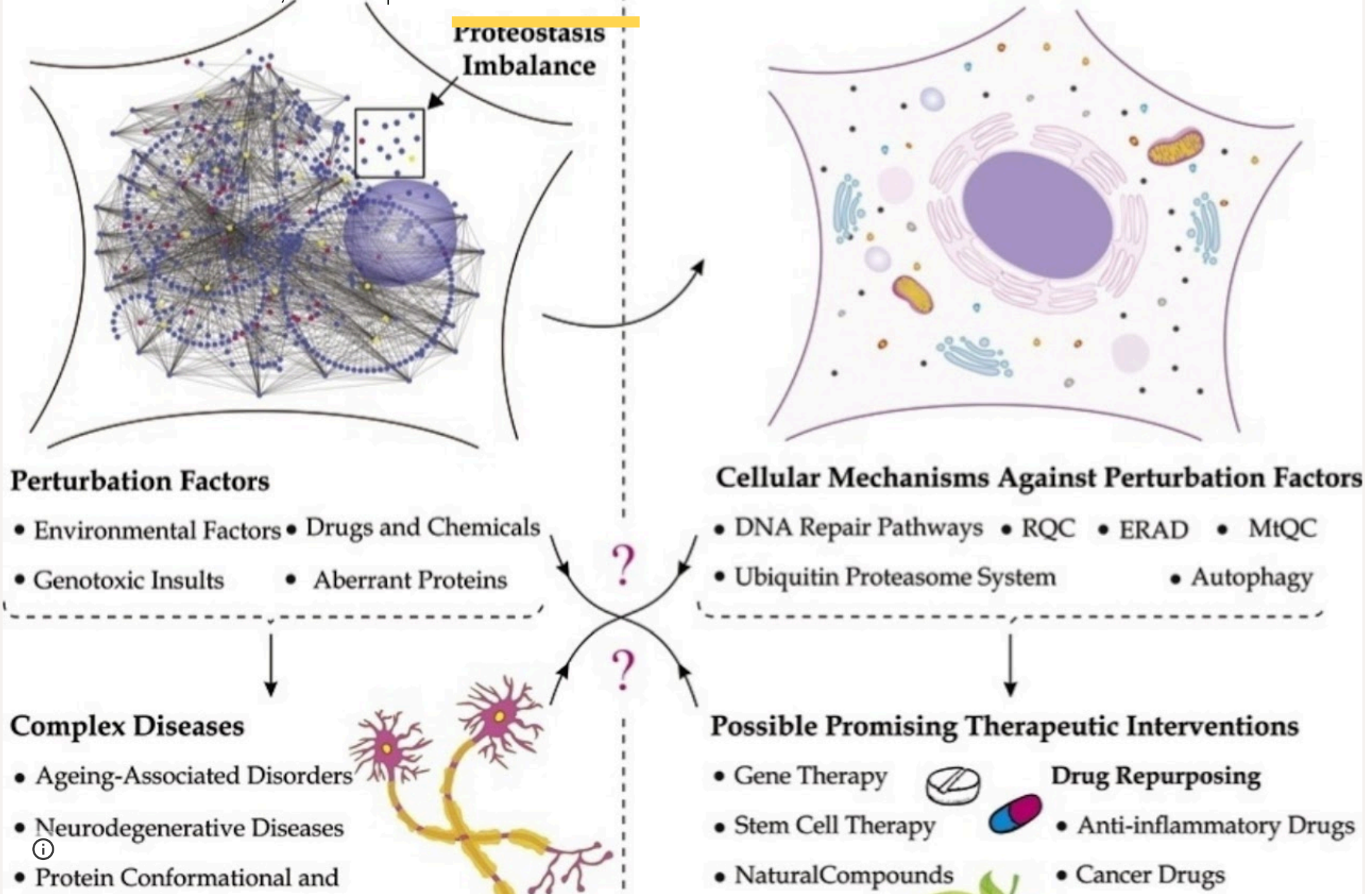
VOLUME 159C ISSN 0301-0082

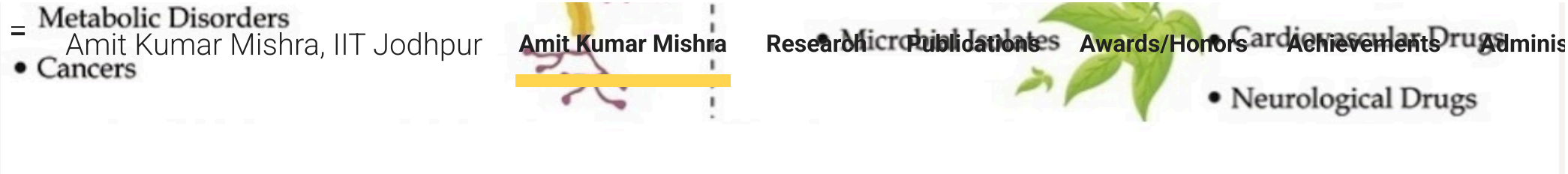
ELSEVIER

PROGRESS IN NEUROBIOLOGY



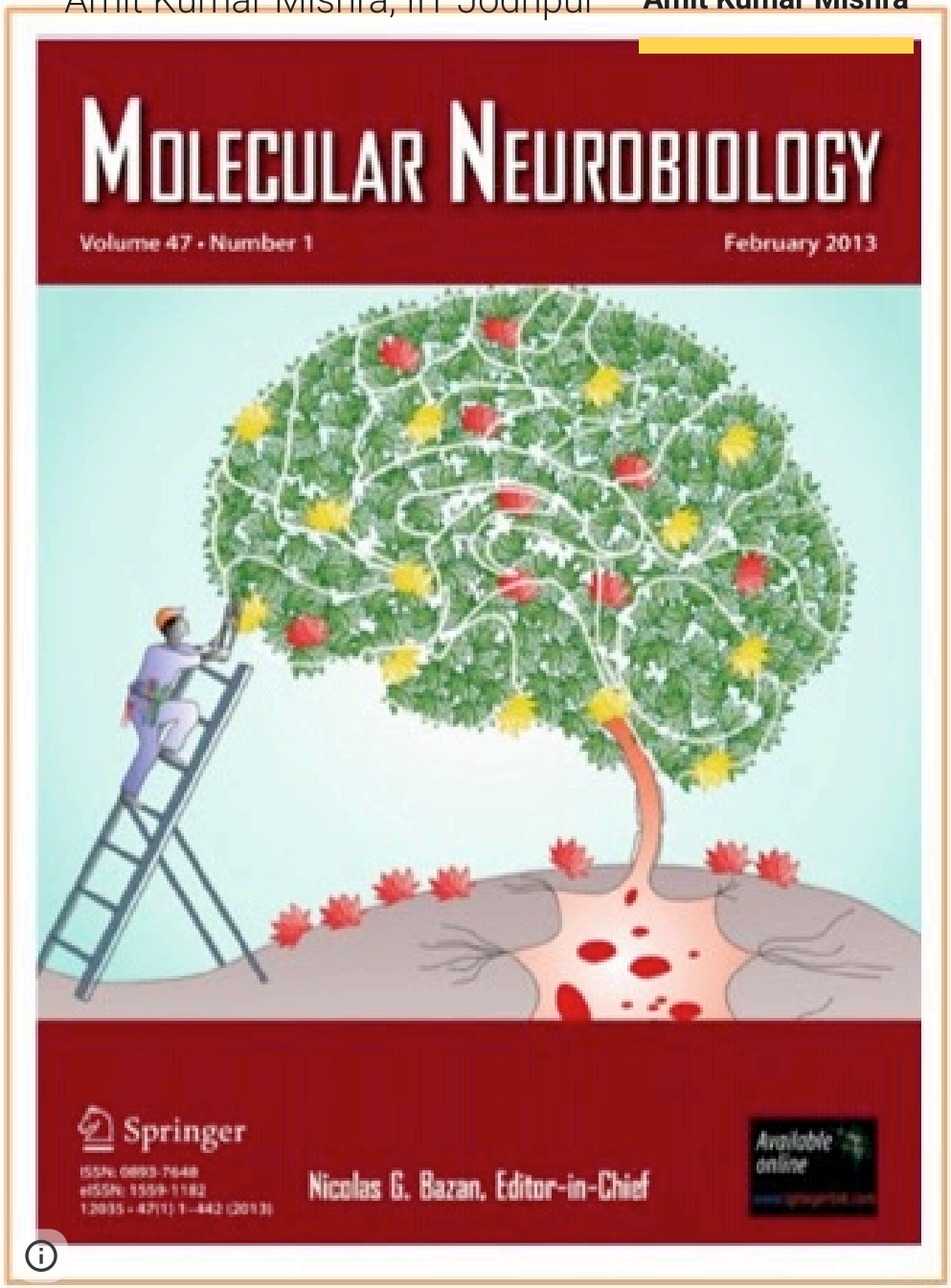


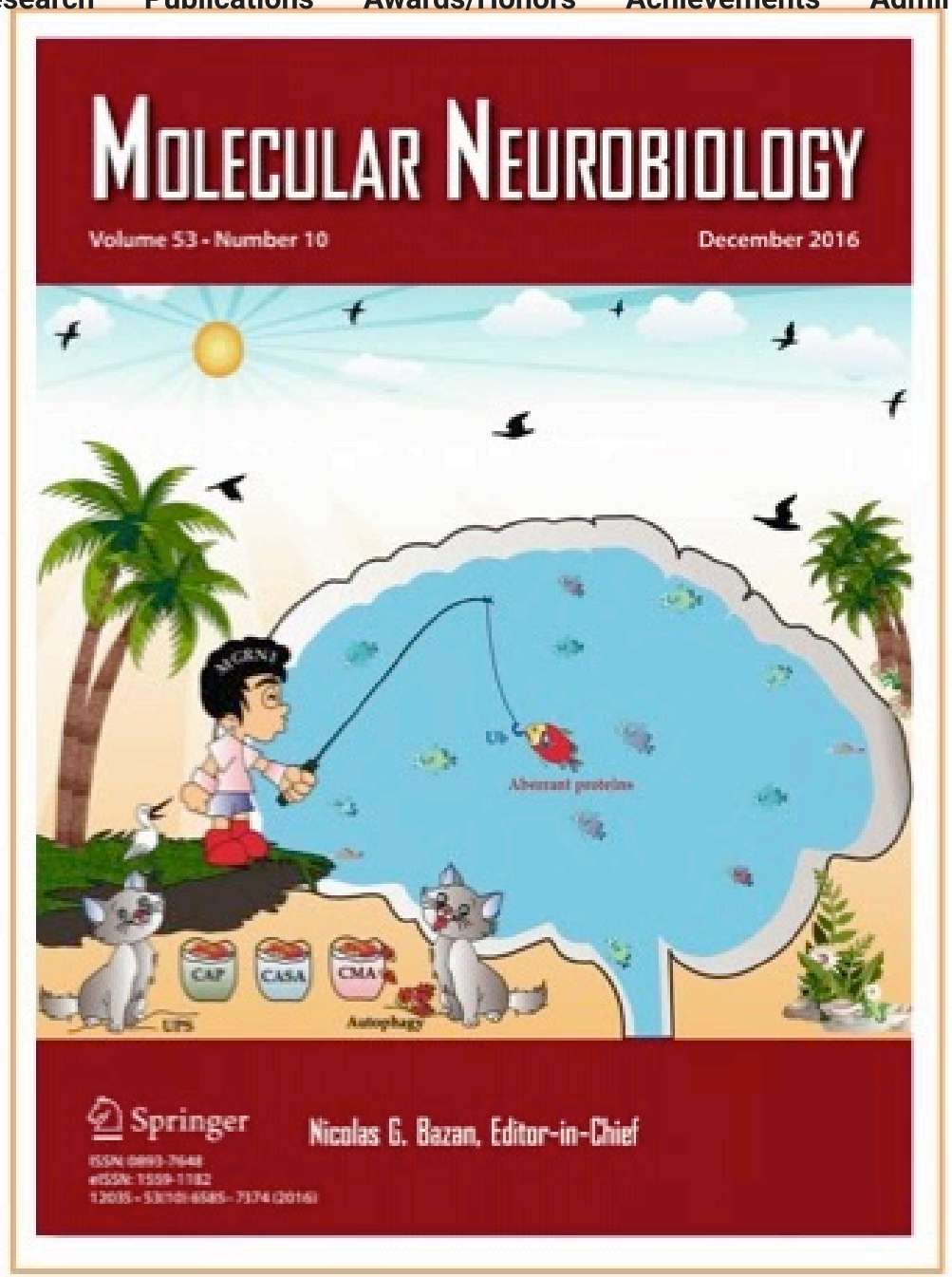

Proteostasis Imbalance
Perturbation Factors
• Environmental Factors
• Drugs and Chemicals
• Genotoxic Insults
• Aberrant Proteins
Cellular Mechanisms Against Perturbation Factors
• DNA Repair Pathways
• RQC
• ERAD
• MtQC
• Ubiquitin Proteasome System
• Autophagy
?
?
Complex Diseases
• Ageing-Associated Disorders
• Neurodegenerative Diseases
• Protein Conformational and
Possible Promising Therapeutic Interventions
• Gene Therapy
• Stem Cell Therapy
• NaturalCompounds
Drug Repurposing
• Anti-inflammatory Drugs
• Cancer Drugs



(Cover Page) Ayeman Amanullah, Arun Upadhyay, Vibhuti Joshi, Ribhav Mishra, Nihar Ranjan Jana and Amit Mishra* (2017) Progressing Neurobiological Strategies Against Proteostasis Failure: Challenges in Neurodegeneration
Progress in Neurobiology DOI:10.1016/j.pneurobio.2017.08.005

Amit has demonstrated that treatment of lanosterol and myricetin can diminish aberrant proteotoxic aggregation and mitigate neurotoxicity. Interestingly he has received the **Best Faculty Research Excellence Award in the presence of the Principal Secretary (Pramod Kumar Mishra) to the Prime Minister of India. Padma Vibhuhassan Prof. P.N Tandon and Shanti Swarup Bhatnagar Prize, Padma Shri Prof. Vijaylakshmi Rabindranath also awarded him S. S. Parmar Foundation Prize, USA under Indian Academy of Neuroscience**. The major histochemical hallmark in the remaining motor neurons of ALS is the intracellular accumulation of ubiquitinated inclusions consisting of insoluble aberrant protein aggregates. However, the molecular pathomechanisms underlying the process have been elusive. For the first time, Amit's lab reported that E6-AP E3 ubiquitin ligase depleted in Amyotrophic Lateral Sclerosis mouse models before Neurodegeneration. His remarkable findings suggest that enhancing the activity of E6-AP ubiquitin ligase might be a viable therapeutic strategy to eliminate mutant SOD1-mediated toxicity in Amyotrophic Lateral Sclerosis. He received a **prestigious Max Planck Society Fellowship by Prof. Ulrich Hartl (Director of Max Planck)**; he has also received a highly competitive Riken Brain Science Institute Fellowship by Presidents of Riken BSI Prof. Masao Ito and Prof. Shunichi Amari.


MOLECULAR NEUROBIOLOGY
Volume 47 • Number 1
February 2013
Springer
Nicolas G. Bazan, Editor-in-Chief
Available online


MOLECULAR NEUROBIOLOGY
Volume 53 - Number 10
December 2016
Aberrant proteins
CAP
CASA
CMA
UPS
Autophagy
Springer
Nicolas G. Bazan, Editor-in-Chief

DBT India awarded him the **Ramalinganswami Fellowship** at an early age and the **IYBA Award, Department of Biotechnology India.** Polyglutamine diseases are a family of inherited neurodegenerative diseases caused by the expansion of CAG repeats within the coding region of target genes. Still, the mechanism(s) by which polyglutamine proteins are ubiquitinated and degraded remains obscure. For the first time, Amit's research demonstrates that MGRN1 E3 ubiquitin-protein ligase is depleted in cells expressing expanded-polyglutamine proteins. MGRN1 interacts with expanded polyglutamine huntingtin and ataxin-3 proteins. His observations suggest that stimulating the activity of MGRN1 ubiquitin ligase might be a potential therapeutic target to eliminate the neurotoxic threat in polyglutamine diseases. His findings were designated among **TOPNOTCH 10 Young Innovative Entries** from ASIA under the BioAsia Drug Discovery and Innovation Programme. He has published high-quality more than a hundred international publications, some of which were selected as Cover Pages of International Journals. **SERB India, DBT India, BRNS/BARC, DST-JSPS, INSA-JSPS, and DST** awarded him crucial research projects. Well-recognized national and International scientific organizations filtered Amit's candidature at multiple levels under different Honors/Awards/fellowships. His research contributions were awarded by different Academics & Research e.g., **CSIR, IITs, MHRD, ICMR, DBT, DST, SERB, BRNS/BARC, INSA, NASI, ISCA, INYAS, IABS, BRSI, NAMS, JSPS, Max Planck, RIKEN, RSC, RSB, NYAS, IGC, and IAN.** Amit's research contribution improves our basic and applied understanding of progress linked with proteome and neurobiological challenges that may provide new basic concepts shortly, based on pharmacological agents linked with impaired proteostasis and neurodegenerative diseases.